

HOSSAM EMAD MOHAMED OSMAN

Embedded Systems / Hardware Engineer

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PROFESSIONAL SUMMARY

Embedded Systems Engineer with 2+ years of hands-on experience in hardware design, PCB layout, firmware development, and production support. Proven ability to independently identify field issues, propose solutions, and deliver improved hardware and firmware revisions end-to-end. Experienced in automotive and industrial IoT systems, 4-layer high-speed PCB design, EMI mitigation, controlled impedance routing, BMS, and RTOS-based firmware — with full product lifecycle ownership from concept through manufacturing and deployment.

TECHNICAL SKILLS

Firmware & Software: Embedded C (production-grade), Python (prototyping & validation), RTOS (FreeRTOS), SMS/MQTT/HTTP, GPS data processing, FOTA, Geofencing algorithms, Embedded debugging, Root-cause analysis

Hardware & PCB: Multi-layer PCB design (up to 4 layers), Stackup planning & controlled impedance routing (USB, CAN, RF), High-speed design, EMI mitigation (ground stitching, return path optimization), RF subsystem design (antenna placement & impedance matching), Power electronics, Battery Management Systems (BMS – 3-cell), Board bring-up, soldering & field debugging, Full manufacturing file preparation (Gerber, BOM, CPL), Fab coordination (JLCPCB)

Protocols & Interfaces: CAN (High-Speed, SWCAN, scanning), USB, UART/SPI/I2C, RS232/RS485, Wi-Fi/BLE, 4G (SIMCom), ADC/GPIO

System & Product Engineering: Full product lifecycle ownership (concept → design → manufacturing → deployment), Reverse engineering & hardware redesign, Mechanical integration & enclosure selection, Automotive telematics & Industrial IoT gateways

Tools: Altium Designer (advanced schematic & layout, actively upskilling), EAGLE, EasyEDA Pro, KiCad, Autodesk Fusion 360, LTspice, JLCPCB manufacturing

PROFESSIONAL EXPERIENCE

Embedded Systems Engineer

Jul 2024 – Present

SEITech Solutions (Acquired by T&S Group) — Smart Village, Cairo

SEIFence – IoT Tracking & Control Device (Hardware & Firmware Version 2)

- Identified hardware, mechanical, and firmware issues in Version 1 during early field deployment across ~200 units.
- Independently proposed, designed, and developed a complete improved Hardware Version 2 (4-layer PCB), reducing costs and enhancing reliability.
- Designed and implemented full RTOS-based production firmware in C: SMS/MQTT/HTTP, GPS processing, FOTA, and geofencing algorithm.
- Resolved GPS antenna stability issues with post-fab PCB fixes (drilled holes + through-hole pads); successfully transitioned 120+ units to production Version 2.
- Provided ongoing aftersales hardware maintenance and debugging for all fielded units.

ForWheelz – Automotive Telematics Device

- Led full hardware development lifecycle for STM32-based automotive telematics ECU with 3 CAN interfaces, USB, RF, and power management.
- Designed 2-board stacked 4-layer impedance-controlled PCB with integrated RF subsystem (GSM antenna, matching network, RF routing constraints).
- Applied EMI mitigation techniques including ground stitching and return path optimization for automotive signal integrity.
- Managed compact enclosure integration under automotive constraints and coordinated full manufacturing package and supplier communication.

- Supported system validation and field optimization through deployment.

E2I – Industrial IoT Gateway & 3-Cell BMS Backup Power Module

- Designed complete 3-cell BMS with controlled DC-DC power path, charging/discharging monitoring, manual assembly, bring-up, and full system validation.
- Reverse-engineered existing Edge Box hardware and redesigned ESP32-S3-based industrial IoT gateway integrating Wi-Fi, BLE, 4G, GPS, RS232/485, ADC, and power management circuits.

Slave Module – Industrial Ethernet & RS485 Remote I/O Controller

- Designed production-ready embedded controller based on STM32F401 Cortex-M4 with Ethernet (ENC28J60), isolated RS485 (MAX3485), USB Type-C, dual EEPROM, relay outputs, and digital I/O.
- Implemented multi-rail power system (12V/5V LM2596 / 3.3V AMS1117) with full ESD, surge, and transient protection.
- Completed full design cycle: schematics, PCB layout, BOM (LCSC), manufacturing files, and production support.

BMS Development (TI & ADI)

- Power circuit design, AFE integration, isoSPI communication, NXP BMS controller, FS85 SBC, and high-voltage system validation.

Tools: EAGLE, EasyEDA Pro, KiCad, Fusion 360, LTspice

Embedded Systems Engineer

Jan 2024 – Feb 2024

Noziqs Medical — Nasr City (Short-term contract engagement)

- Developed embedded firmware for operating tables, improving system functionality by 30%.
- Reduced device downtime by 20% via board-level troubleshooting and root-cause analysis.

EDUCATION

Bachelor of Electronics and Control Engineering

Sep 2017 – Jul 2022

Menoufia University | Grade: Very Good

Note: Following graduation, pursued the NTI Automotive Embedded Systems Diploma (Feb–Jul 2024) before transitioning to industry.

CERTIFICATIONS & TRAINING

Crash Course Electronics & PCB Design — Udemy

Jan 2026

112 hours covering AC/DC analysis, analog/digital electronics, PCB design flow: schematic → layout → fabrication → bring-up.

Automotive Embedded Systems Diploma — NTI, New Capital

Feb 2024 – Jul 2024

Embedded C, RTOS, STM32 & AVR, CAN/LIN/UART/SPI/I2C, bootloader/FOTA, automotive testing, project management.

Hardware Design Mentorship (Ongoing)

Professional PCB design (Altium Designer), ESP32-S3 & RP2040 dev boards, advanced grounding and layout practices.

LANGUAGES

English — Professional Proficiency